Names: Michelle Tan, Grayson Schultz, Christopher Pan

Github Repo: https://github.com/UC-Berkeley-I-School/200-FA26-EDA-Semiconductors

Mode of Communication: Slack

Regular meeting time: Saturdays 2:00 PM Pacific Time

Dataset:
https://www.kaggle.com/datasets/sergionefedov/global-semiconductor-industry-2010-2026

This dataset contains a variety of data about the global semiconductor industry from 2010-2026. This includes data on the financials of semiconductor companies, the kinds of chips that are being sold and the amount, as well as data on the overall performance level of the semiconductors and how they have changed over time. The collection methodology explains that the financial data is anchored to public 10-K forms that are required by the U.S Securities and Exchange Commision for publicly traded companies. Chip prices and chip market data are pulled from DRAMAeXchange/TrendForce and Omdia, which are the industry-standard for semiconductor market intelligence.

This dataset contains 5 csv files:

- ai_chip_market.csv
- chip_companies_financials.csv
- chip_prices.csv
- export_contraols.csv
- fab_capacity.csv

Interesting Variables / Columns:

- Revenue: Shows performance and level of activity of semiconductor companies
- CapEx: Indicates level of investment that semiconductor companies are making, and how much growth the companies believe there will be in the future
- Estimated Shipment Units: Indicates level of demand for chips
- Chip prices: Shows level of demand for chips, and how much people are willing to pay, most likely changed massively over the past few years as large amount of wealthy buyers have entered the market
- Product: What kind of chip, indicates what kinds of semiconductor chips, and therefore what kind of computation, is in demand
- Export controls: Which countries are not allowed to import chips, helps show global semiconductor competition and impact of trade wars

Guiding Questions:

- How have the amount of chips sold, and the type of chips sold, changed over this decade? (2020-2026)
- How have export controls impacted chip supply chains and revenue?
- How did AI impact semiconductor revenue?

Supplemental Datasets:

- Yahoo finance dataset - provides stock information for semiconductor companies
- Ramp AI Index dataset - Tracks AI spending among Ramp clients
- Cloud compute dataset - Find a dataset that measures usage of cloud computing services - Chris

Timeline of Milestones:

- November 13th - Refined guiding questions, supplemental datasets, and define members' scope and responsibilities
- November 20th - Clean data, formatted well for analysis
- November 27th - Data analysed, graphs built
- December 4th - Write up finished Presentation and Jupiter note book complete with everyone aligned on what to say during presentation
- December 7th - Presentation Day